AI-Assisted Sales Performance Analysis

Management Analysis and Business Insights

1. Objective

The objective of this analysis is to evaluate employee sales performance using monthly sales, individual targets, attendance, customer complaints, and order data. Microsoft Excel was used to calculate and review key performance metrics, while Generative AI was used to assist with formula development, business analysis, management summaries, and recommendations. All AI-generated outputs were independently reviewed against the Excel calculations and source dataset before being used.

2. Dataset

The practice dataset contains information for 10 employees across 5 departments and 4 regions. The dataset includes employee, department, region, product, monthly sales, target, attendance, customer complaints, and order information. The analysis focuses primarily on Sales employees because recorded sales and targets are associated with the Sales team, while other departments have zero recorded sales and targets in this dataset.

3. Key Findings

- **Finding 1-** Sales Performance: The team recorded total monthly sales of ₹370,000 against a total target of ₹370,000, resulting in 100% overall achievement. Priya recorded the highest Sales-team sales at ₹91,000, while Arjun recorded the lowest Sales-team sales at ₹54,000.
- **Finding 2-** Individual Target Achievement: The Sales team achieved 100% of its combined monthly target despite Rahul and Arjun performing below their individual targets. This indicates that individual performance varied while the combined team target was achieved.
- **Finding 3-** Attendance and Customer Complaints: Average employee attendance was 92.1%, while four employees recording attendance below 90%. Among Sales employees, Arjun had the lowest attendance and the highest number of customer complaints. These observations identify areas for further investigation, but the dataset does not establish a causal relationship between attendance, complaints, and sales performance.

4. AI Assistance

Generative AI was used as an analytical assistant throughout the project. It supported Excel formula development, helped generate management questions, converted verified findings into a concise management summary, and assisted in developing practical management recommendations. AI was used to support the analysis rather than replace independent judgement or Excel-based verification.

5. Verification

All important AI-generated outputs were independently checked against the Excel calculations and source dataset. Numerical results were verified before inclusion in the analysis. AI-generated statements were also reviewed to identify unsupported assumptions, fabricated information, and causal claims that were not established by the dataset. Only findings supported by the available data were retained.

6. Management Recommendations

Recommendation 1- Review below-target employees: Management should review Rahul's and Arjun's performance against their individual targets and identify specific areas requiring attention. Further investigation is needed to determine the reasons for their individual sales shortfalls.

Recommendation 2- Review Attendance below 90%: Management should review attendance patterns for the four employees with attendance below 90%, with particular attention to Arjun, who has the lowest attendance among Sales employees. Further investigation is needed to understand the reasons for these attendance levels.

Recommendation 3- Review Arjun's sales and customer-complaint performance: Management should conduct a focused review of Arjun's sales, attendance, and customer-complaint figures because he has the lowest Sales-team sales, lowest Sales-team attendance, and highest number of customer complaints among Sales employees. Further investigation is required before concluding whether these measures are related.

7. Conclusion

The analysis provides a snapshot of sales performance, target achievement, attendance, customer complaints, and employee-level results. The Sales team achieved 100% of its combined target, while differences in individual performance identify areas for management review. The finding should be used as evidence for further investigation rather than as definitive explanations of performance. Additional information and analysis over a longer period would be required to establish broader performance patterns or relationships.